

ANSWER KEY

GATE CS — GATE CS 2026 Trend-Based Mock Paper
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Q.No.	Type	Answer	Marks
1	3MCQ	—	3
2	2MCQ	—	2
3	2MCQ	—	2
4	2MCQ	—	2
5	3MCQ	—	3
6	2MSQ	—	2
7	2MCQ	—	2
8	2MCQ	—	2
9	2MSQ	B	2
10	2MCQ	B	2
11	2NAT	C	2
12	2NAT	B	2
13	2MSQ	C	2
14	1MCQ	B	1
15	1MCQ	B	1
16	1NAT	B	1
17	1MCQ	C	1
18	1MCQ	A	1
19	1NAT	C	1
20	1NAT	C	1
21	1MCQ	C	1
22	1MCQ	B	1
23	1MCQ	B	1
24	2MSQ	C	2
25	1NAT	B	1
26	1NAT	A	1
27	1MCQ	B	1
28	2MCQ	C	2
29	1MCQ	B	1
30	1MCQ	A	1
31	1MCQ	C	1
32	1NAT	B	1
33	1MCQ	C	1
34	1MCQ	B	1
35	1MCQ	B	1
36	1MCQ	C	1

37	1MCQ	B	1
38	1MCQ	B	1
39	1MCQ	B	1
40	1NAT	B	1
41	1MCQ	A	1
42	1MCQ	B	1
43	1NAT	B	1
44	1NAT	B	1
45	1MCQ	C	1
46	1MCQ	A	1
47	2NAT	—	2
48	2NAT	—	2
49	2MSQ	—	2
50	2MSQ	—	2
51	2MSQ	—	2
52	2MCQ	—	2
53	2MCQ	—	2
54	2MCQ	—	2
55	2MCQ	—	2
56	2MCQ	—	2
57	2MCQ	—	2
58	2NAT	—	2
59	1NAT	A	1
60	2NAT	—	2
61	2NAT	—	2
62	2NAT	—	2
63	2MCQ	—	2
64	2MCQ	—	2
65	2MCQ	—	2

Q1. General Aptitude | 3 marks | 3MCQ

Let 1 and 2 denote two arbitrary prime numbers. Which one of the following statements is correct for all values of 1 and...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q5

Q2. General Aptitude | 2 marks | 2MCQ

The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q4

Q3. General Aptitude | 2 marks | 2MCQ

Consider the relationships among P, Q, R, S, and T: $\ddot{y}y$ P is the brother of Q. $\ddot{y}y$ S is the daughter of Q. $\ddot{y}y$ T is the sister..

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q4

Q4. General Aptitude | 2 marks | 2MCQ

Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered. Select the...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q1

Q5. General Aptitude | 3 marks | 3MCQ

According to the map shown in the figure, which one of the following statements is correct? Note: The figure shown is re...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q5

Q6. General Aptitude | 2 marks | 2MSQ

The average marks obtained by a class in an examination were calculated as 30.8. However, while checking the marks enter...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q3

Q7. General Aptitude | 2 marks | 2MCQ

Ravi had _____ younger brother who taught at _____ university. He was widely regarded as _____ honorable man. Select ...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q1

Q8. General Aptitude | 2 marks | 2MCQ

Two wizards try to create a spell using all the four elements, water, air, fire, and earth. For this, they decide to mix...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q2

Q9. General Aptitude | 2 marks | 2MSQ

Synonym of 'ubiquitous':

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q10

Q10. Engineering Mathematics | 2 marks | 2MCQ
Derivative of x^3 at $x=2$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q1

Q11. Engineering Mathematics | 2 marks | 2NAT
Let A be a 3×3 matrix with eigenvalues 1, 2, and 3. Then the trace of A^2 is:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q1

Q12. Engineering Mathematics | 2 marks | 2NAT
Mean of binomial $n=10, p=0.5$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q1

Q13. Engineering Mathematics | 2 marks | 2MSQ
Probability of at least one head in 3 coin tosses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q11

Q14. Engineering Mathematics | 1 marks | 1MCQ
Determinant of 2×2 identity:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q1

Q15. Programming and Data Structures | 1 marks | 1MCQ
Infix to postfix uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q6

Q16. Programming and Data Structures | 1 marks | 1NAT
What is the time complexity of inserting at the beginning of a linked list?

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q3

Q17. Programming and Data Structures | 1 marks | 1MCQ
Avg hash lookup:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q3

Q18. Programming and Data Structures | 1 marks | 1MCQ
Inorder traversal of BST gives:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q3

Q19. Algorithms | 1 marks | 1NAT
LCS uses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2023 CS Session 1 Q7

Q20. Algorithms | 1 marks | 1NAT
Merge sort space:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q7

Q21. Algorithms | 1 marks | 1MCQ
Kruskal uses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 CS Session 1 Q2

Q22. Algorithms | 1 marks | 1MCQ
Time complexity of binary search:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q11

Q23. Algorithms | 1 marks | 1MCQ
BFS uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q7

Q24. Algorithms | 2 marks | 2MSQ
The time complexity of Dijkstra's algorithm using a binary heap is:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q7

Q25. Algorithms | 1 marks | 1NAT
Worst-case time complexity of merge sort:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q12

Q26. Operating Systems | 1 marks | 1NAT
Round-robin uses:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q4

Q27. Operating Systems | 1 marks | 1MCQ
Banker's algorithm is for:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q4

Q28. Operating Systems | 2 marks | 2MCQ
*Using 3 frames, which page replacement algorithm would not cause any page fault for reference string
0,1,2,3,0,1,4,0,1,2...*

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q8

Q29. Operating Systems | 1 marks | 1MCQ
Inode contains:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q4

Q30. Operating Systems | 1 marks | 1MCQ
TLB stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 CS Session 1 Q9

Q31. Operating Systems | 1 marks | 1MCQ
Which of the following is NOT a necessary condition for deadlock?

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q4

Q32. Operating Systems | 1 marks | 1NAT
Semaphore with value 1 is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q8

Q33. Computer Networks | 1 marks | 1MCQ
Asymmetric encryption:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 CS Session 1 Q5

Q34. Computer Networks | 1 marks | 1MCQ
CSMA/CD is used in:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q13

Q35. Computer Networks | 1 marks | 1MCQ
Protocol to find MAC from IP:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q6

Q36. Computer Networks | 1 marks | 1MCQ
Which protocol operates at the application layer of the TCP/IP model?

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q6

Q37. Computer Networks | 1 marks | 1MCQ
TCP connection establishment uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q14

Q38. Computer Networks | 1 marks | 1MCQ
OSPF is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q6

Q39. Digital Logic | 1 marks | 1MCQ
The number of Boolean functions of n variables is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q2

Q40. Digital Logic | 1 marks | 1NAT
Full adder has how many inputs:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q2

Q41. Digital Logic | 1 marks | 1MCQ
Flip-flop stores:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 CS Session 1 Q2

Q42. Theory of Computation | 1 marks | 1MCQ
Halting problem:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q8

Q43. Theory of Computation | 1 marks | 1NAT
PDA recognizes:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q14

Q44. Theory of Computation | 1 marks | 1NAT
Which language is regular?

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q13

Q45. Compiler Design | 1 marks | 1MCQ
3-address code operands:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q9

Q46. Compiler Design | 1 marks | 1MCQ
Lexical analyzer outputs:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q15

Q47. CS | 2 marks | 2NAT
8 bytes in 1 cycle from a device to main memory through cycle stealing at regular intervals. Which one of the following ...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q11

Q48. CS | 2 marks | 2NAT
Consider the pushdown automaton (PDA) P below, which runs on the input alphabet $\{a, b\}$, has stack alphabet $\{A\}$, and ...

Answer: See explanation

Source: Adapted from GATE 2023 CS Session 1 Q40

Q49. CS | 2 marks | 2MSQ
Let f be a 3-variable Boolean function that produces output as '1' when at least two of the input variables are '1'. Which...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q24

Q50. CS | 2 marks | 2MSQ
The pseudocode of a function $fun()$ is given below: $fun(int A[0,...,n-1])$ { for $i=0$ to $n-2$ for $j=0$ to $n-i-2$ if $(A[i]>A[j+1]...$

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q33

Q51. CS | 2 marks | 2MSQ

Consider a Boolean expression given by $(,) = (3, 5, 6, 7)$. Which of the following statements is/are CORRECT?

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 1 Q47

Q52. CS | 2 marks | 2MCQ

Consider a disk with the following specifications: rotation speed of 6000 RPM, average seek time of 5 milliseconds, 500 ...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q53

Q53. CS | 2 marks | 2MCQ

A square paper, shown in figure (I), is folded along the dotted lines as shown in the figures (II) and (III). Then a few...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q9

Q54. CS | 2 marks | 2MCQ

Let $R1$ and $R2$ be two 4-bit registers that store numbers in 2's complement form. For the operation $R1 + R2$, which one of th...

Answer: See explanation

Source: Adapted from GATE 2022 CS Session 1 Q18

Q55. CS | 2 marks | 2MCQ

Consider a digital display system (DDS) shown in the figure that displays the contents of register X. A 16-bit code word...

Answer: See explanation

Source: Adapted from GATE 2022 CS Session 1 Q40

Q56. General Aptitude | 2 marks | 2MCQ

Let 1 and 2 denote two arbitrary prime numbers. Which one of the following statements is correct for all values of 1 and...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q5

Q57. CS | 2 marks | 2MCQ

$= \{0, 1, 2, 3, \dots\}$ is the set of non-negative integers. Let be the set of functions from to itself. For any two functi...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q49

Q58. CS | 2 marks | 2NAT

An instruction format has the following structure: Instruction Number: Opcode destination reg, source reg-1, source reg...

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q31

Q59. Programming and Data Structures | 1 marks | 1NAT

Inorder traversal of BST gives:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q3

Q60. CS | 2 marks | 2NAT

Consider a demand paging system with three frames, and the following page reference string: 1 2 3 4 5 4 1 6 4 5 1 3 2. T...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q47

Q61. CS | 2 marks | 2NAT

Let $S = \{1, 2, 3, 4\}$. For $s \in S$, let $s()$ be the product of symbols in modulo 7. We take $s() = 1$, where s is the null string. For examp...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 2 Q60

Q62. CS | 2 marks | 2NAT

The maximum value of k such that the edge between the nodes B and C is included in every minimum spanning tree of the given graph is...

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q64

Q63. CS | 2 marks | 2MCQ

In the context of owner and weak entity sets in the ER (Entity-Relationship) data model, which one of the following statements is true?

Answer: See explanation

Source: Adapted from GATE 2024 CS Session 2 Q20

Q64. CS | 2 marks | 2MCQ

"I put the brown paper in my pocket along with the chalks, and possibly other things. I suppose every one must have referred to it..."

Answer: See explanation

Source: Adapted from GATE 2025 CS Session 1 Q6

Q65. CS | 2 marks | 2MCQ

The value of the definite integral $\int_0^1 (4x^2y - z^3) dz dy dx$ is $-3/2 - 1/2$. (Rounded off to the nearest integer) CS GATE 2023...

Answer: See explanation

Source: Adapted from GATE 2023 CS Session 1 Q31